

ARTI 404 - Image Processing

Lab 3 Report: Image Manipulations using OpenCV

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Course: ARTI 404 - Image Processing

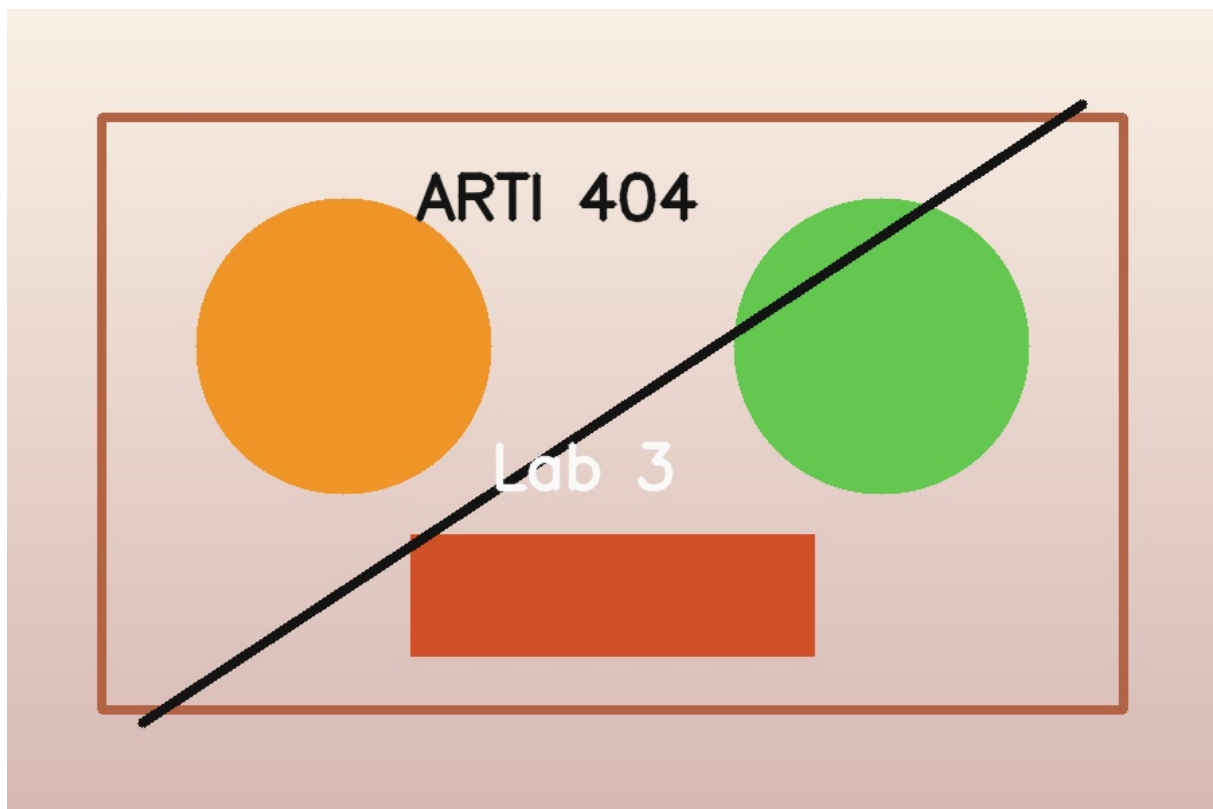
Lab: 3

Objectives

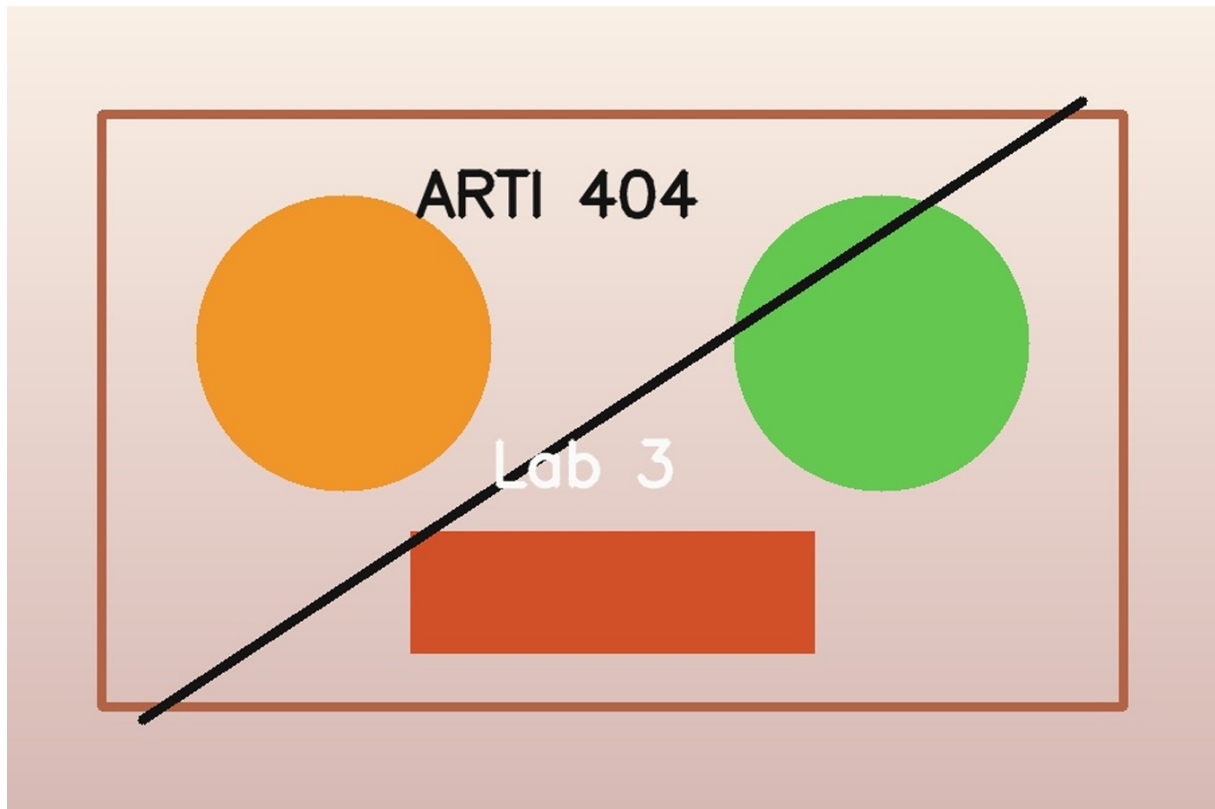
Apply the required geometric and intensity transformations to a digital image using Python and OpenCV.

Task	Operation	Parameter
Task 1.1	Resize	150%
Task 1.2	Rotation	120 degrees
Task 1.3	Shear	X shear = 0.35
Task 2.1	Negative	$255 - r$
Task 2.2	Log	$c = 255 / \log(1 + \max(r))$
Task 2.3	Power-law	$\gamma = 0.5$

Input Image



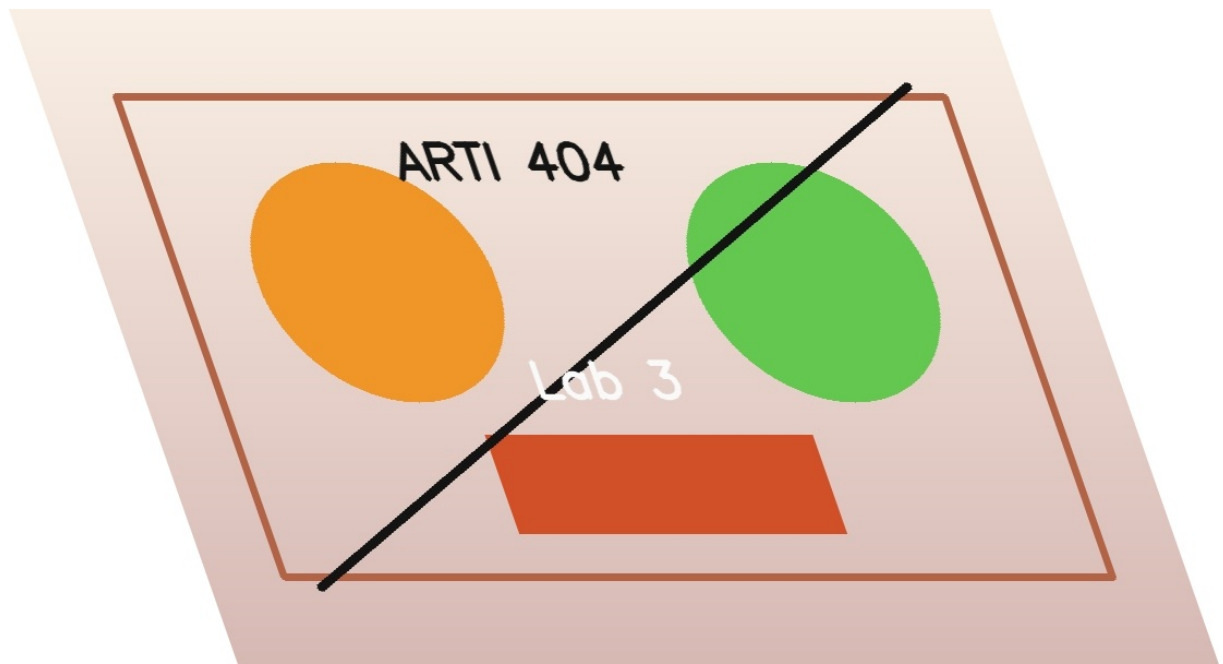
Task 1.1 - Increased Image Size



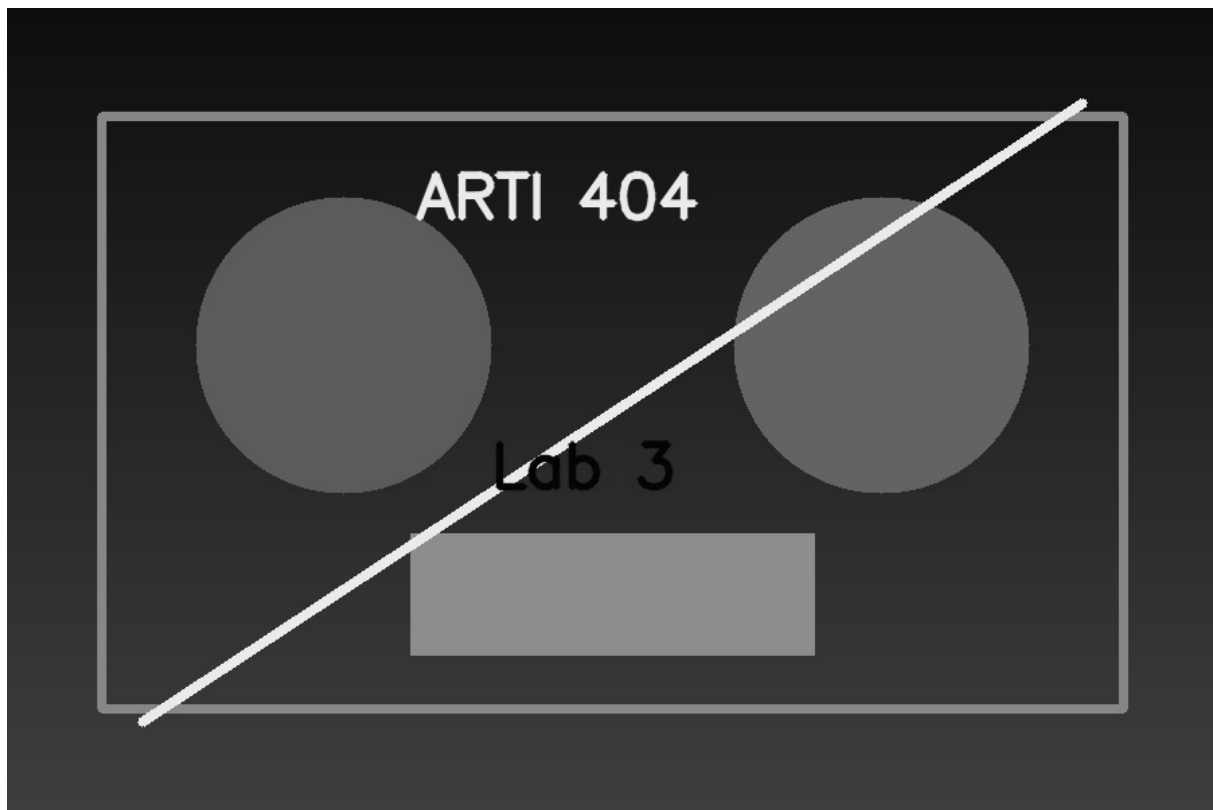
Task 1.2 - Rotation by 120 Degrees



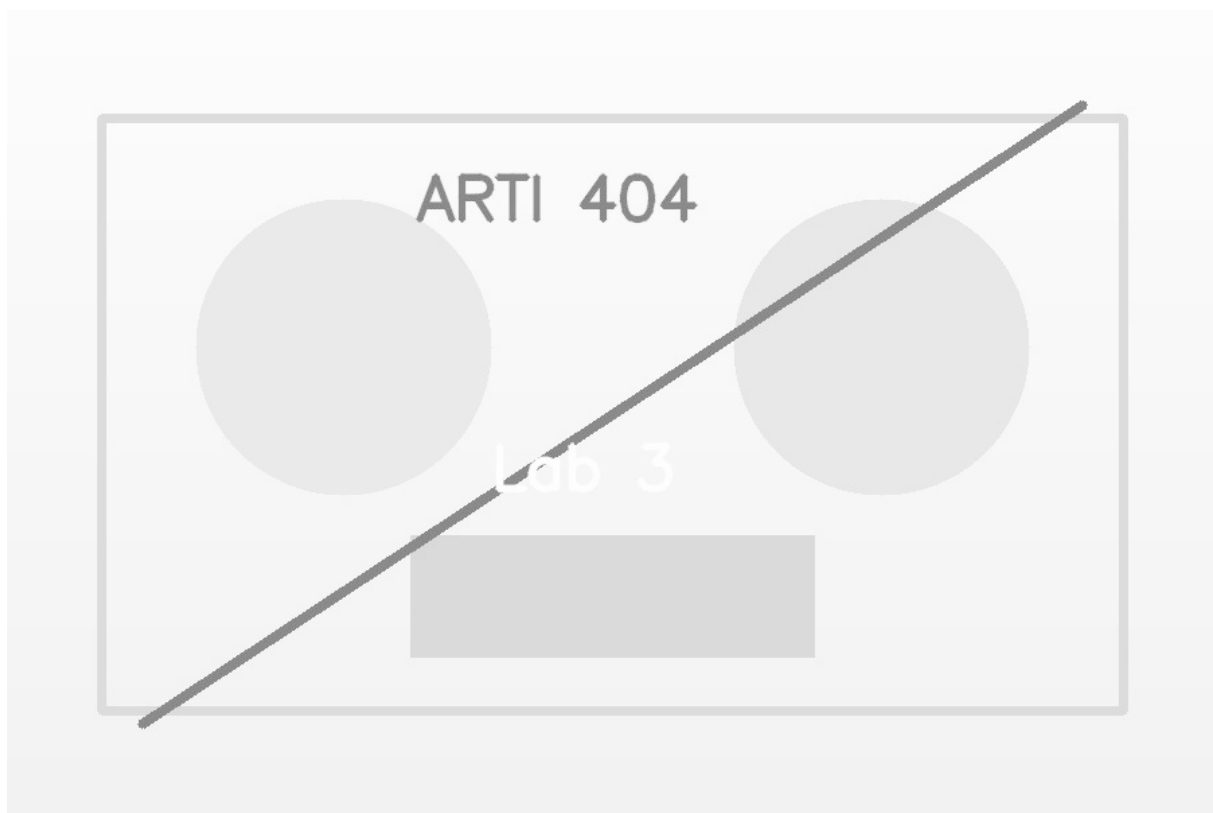
Task 1.3 - Shear Operation



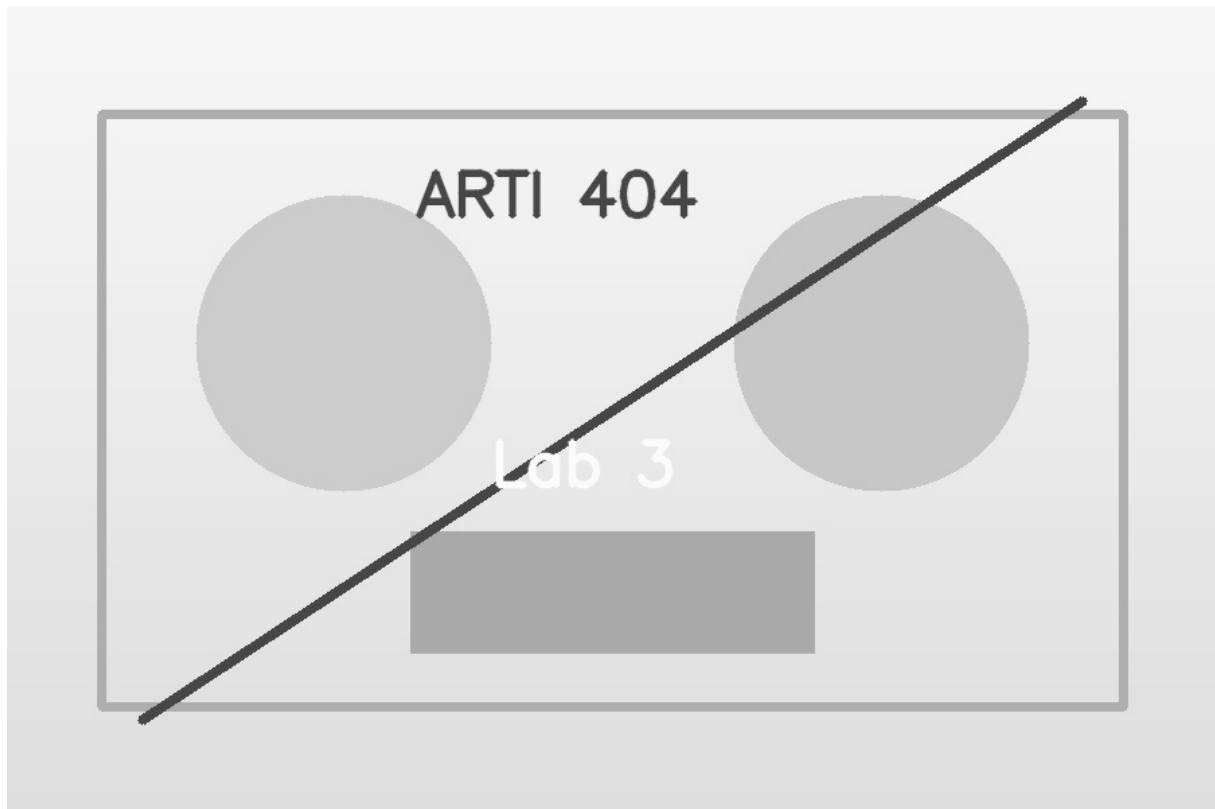
Task 2.1 - Negative Image



Task 2.2 - Log Transformation



Task 2.3 - Power-Law Transformation (gamma = 0.5)



Observations

The geometric operations change the spatial arrangement or size of the image. The intensity operations change pixel values while preserving the image dimensions. The negative reverses brightness, the logarithmic transformation expands lower intensities, and the gamma transformation with $\gamma = 0.5$ makes darker regions brighter and easier to inspect.

Conclusion

All transformations requested in Lab 3 were implemented with OpenCV and saved as separate image files. The notebook contains the complete code and visual outputs.